



Effect of motion trajectory on the aerodynamic performance of a flapping airfoil



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HIGHLIGHTS

- A surging motion is added to the traditional motion of a flapping airfoil.
- Thrust and lift forces can be improved under specific condition.
- Two-circle motion model is helpful for propulsive efficiency enhancement.

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ABSTRACT

In this paper, the influence of the motion trajectory on the aerodynamic performance of a flapping airfoil is numerically investigated. The airfoil synchronously performs a rotating motion (pitch), a horizontal motion (surge) and a vertical motion (plunge). The motion trajectory can be modified by changing the relative frequency of surging motion (k), the amplitude of surging motion (l_m) and the phase difference between pitch and surge (φ). To perform the numerical simulations, a NACA0012 airfoil is employed. A low Reynolds number ($Re = 500$) is selected for small creatures and flapping MAVs. As compared with the traditional motion, it is observed that the figure-of-eight motion trajectory has a promotion for the thrust and propulsive efficiency, and the lift is improved when the relative surging frequency k is an odd number with the specified phase difference. In addition, the phenomenon of multiple vortices and resultant vortex capture appears when the surging motion is added, which is helpful for the improvement of thrust.

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1. Introduction

The development of micro aerial vehicles (MAVs) and nano aerial vehicles (NAVs) has attracted a great number of researches recently, because of their importance in both theory development and practical applications. Actually, the inventions of MAVs and NAVs are inspired from insects, birds and swimming animals that utilize flapping wings motion as a mode of propulsion in air or water (Rozhdestvenshy and Ryzhov, 2003). Therefore, the use of flapping wings systems shows great promise and prospect in locomotion applications. The early studies of flapping wings for propulsion were mainly

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performed through theoretical analysis (Garrrick, 1936). Later, more and more investigations on flapping wings systems were completed by both experiments and numerical simulations (Jones et al., 1998; Triantafyllou et al., 2004). However, owing to the complexity of the flow field and fluid–structure interactions involved in the flapping motion, there is still tremendous room for exploring how to make such systems more efficient.

Up to now, to achieve the enhancement of both thrust and the corresponding propulsive efficiency of flapping wings systems, a great deal of effort has been devoted and a large number of techniques have been employed. Tian et al. (2012) put forward a novel strategy by introducing a traveling-wave motion into the surface of a foil. It was found that the larger propulsive efficiency could be achieved as compared with most aquatic animals, which may lead to an important design capability for underwater vehicles. Mivehchi et al. (2016) experimentally investigated the ground effect and found that the distance between the ground and the flapping foil has a significant impact on the thrust force. Meanwhile, the propulsive efficiency is increased slightly when the foil operates near the ground. Broering and Lian (2015) numerically studied the vortex interactions of tandem flapping wings. It was indicated that zero phase lag could increase the size of the leading edge vortex (LEV) generated by the hindwing and the resulting thrust production. Lee and Lee (2013), Lee et al. (2015) analyzed the mechanism of a flapping foil moving with a propulsive velocity. They announced that the symmetric rotation produced a better propulsion velocity with less fluctuation as compared with the advanced rotation and delay rotation. Lu et al. (2013) concluded that large pitching amplitude generates much more thrust than the lower pitching amplitude. However, when the pitching amplitude exceeds some specified limit, the thrust increases slowly and the propulsive efficiency declines noticeably. Tian et al. (2016) studied the effect of pitch–pivot-point location, and they pointed out that the rotating axis position affects the evolution of unsteady wake vortices and propulsive performance. It can be explained that by moving the pitch–pivot-point of pitching foil can be considered as adding a plunging motion to the original pitching motion. Ashraf et al. (2011) proved that the rate of thrust improvement is much better when the plunging amplitude is lower, and the performance of thin airfoils outperforms thick airfoils.

Besides the above mentioned researches, many other studies have been focused on the lift enhancement of the flapping foil. Mantia and Dabnichki (2012) illustrated that the lift force generated by flapping wing increases sharply when the added mass contribution is considered. Young and Lai (2004) demonstrated that leading-edge effects are of great importance in determining the forces, whereas the trailing-edge effects influence primarily on wake structures. They also found that the lift force strongly depends on oscillation frequency. Sarkar et al. (2013) evaluated the effects of asymmetric kinematics on the lift enhancement capabilities of a flapping foil during hover. The normal hover shows a significant increase in the average lift value compared with the sinusoidal hover, and a large amount of lift can be generated during the faster stroke for the sinusoidal hover. Amiralaie et al. (2010) revealed that unsteady parameters such as oscillating amplitude, reduced frequency and Reynolds number play a great role in the development of the maximum lift value, strength and numbers of the generated vortices, and even the surrounding flow structures.

In addition, some attention has also been paid to the effect of camber and flexible airfoil on the aerodynamic performance. Hoke et al. (2015) indicated that varying camber provides little improvement in thrust and propulsive efficiency, but the interactions between LEV and foil curvature result in great propulsive forces. Jaworski and Gordnier (2015) concluded that the maximum of thrust and propulsive efficiency are obtained when the membrane pressure is changed, and it was found that the vortices can interact favorably with the local membrane camber to enhance the propulsive force. Tian et al. (2013) found that the passive pitching motion of a flexible wing can significantly increases thrust and maintain lift at the same level or increase it synchronously.

As discussed above, great progress has been made in the propulsive characteristics of flapping wings systems. However, up to date, there is still limited attention paid to the effect of motion trajectory on the aerodynamic performance of flapping foil. Only few examples can be found in the literature. Sarkar and Venkatraman (2005) studied the effect of three motion modes for propulsive performance: asymmetric motion, sinusoidal motion and constant heave rate oscillations motion. They pointed out that both asymmetric motion and a train of sinusoidal pulses motion give better thrust and propulsive efficiency in comparison to sinusoidal motion. However, constant rate heave motion does not compare favorably with traditional harmonic motion. Amiralaie et al. (2011) performed a modified figure-of-eight-like flapping pattern by using a thin ellipsoidal geometry. It was found that the parameters change the instantaneous force coefficients quantitatively and qualitatively. Esfahani et al. (2015) investigated the effect of elliptical motion trajectory on the flapping airfoil. It was noted that motion trajectories change both the effective angle of attack and the vortex shedding pattern. As a result, they influence significantly the aerodynamic and propulsive performance.

Therefore, to further explore mechanics of a flapping foil, this work continues to investigate the effect of motion trajectory of a flapping airfoil on the propulsive performance and lift enhancement as well. A NACA0012 airfoil, which is placed in a two-dimensional laminar flow, synchronously executes a combined periodic pitching, surging (forwards/backwards) and plunging (upwards/downwards) motion. After selecting the Reynolds number and plunging amplitude, the effects of pitching amplitude, surging amplitude, location of pitch–pivot-point, phase difference between pitch and surge as well as surging frequency on the aerodynamic performance are numerically examined in detail. Based on the numerical results obtained, the thrust and lift coefficients, propulsive efficiency as well as the vorticity and pressure coefficient contours are meticulously analyzed.

Table 1
All parameters used in this work.

Reynolds number	Re	(500)
Reduced frequency	St	[0.2, 0.22, 0.24, 0.26, 0.28, 0.3]
Amplitude of plunging motion	h_m	[c]
Amplitude of pitching motion	θ_m	[15°, 30°, 45°, 60°]
Amplitude of surging motion	l_m	[0.05 c , 0.1 c , 0.15 c , 0.2 c]
Location of pitch–pivot–point	x_p	[$c/4$, $c/3$, $c/2$, $2c/3$]
Adjustable parameter	k	[0, 1, 2, 3, 4]
Phase difference	φ	[0°–315°] (interval is 45°)

2. Problem description and methodology

2.1. Problem description

In this study, the defined traditional kinematics of a NACA0012 airfoil includes two motions: (a) vertical motion (plunge) and (b) rotational motion (pitch). The principal plunging motion and the pitching motion can be expressed as follows:

$$h(t) = h_m \cos(2\pi ft), \quad (1)$$

$$\theta(t) = \theta_m \sin(2\pi ft), \quad (2)$$

where $h(t)$ and $\theta(t)$ represent the instantaneous displacement of the pitch–pivot–point in vertical direction and the instantaneous rotating angle at time t , respectively. h_m and θ_m are the corresponding plunging and pitching amplitudes. f is the oscillating frequency. In this work, in order to focus on the other parameters, the plunging amplitude is fixed at $h_m = c$, c is the chord length of airfoil. Moreover, the pitching amplitude is selected at four angles, i.e., $\theta_m = 15^\circ, 30^\circ, 45^\circ$ and 60° , and the location of pitch–pivot–point also has four options, i.e., $x_p = c/4, c/3, c/2$ and $2c/3$.

On the other hand, animals such as penguins and turtles may add a more similar motion to the horizontal direction of their wings to enhance the propulsive performance (Esfahani et al., 2015; Licht et al., 2010). As a consequence, the resultant kinematics may produce different motion trajectory, such as figure-of-eight and double-eight trajectories (Lehmann and Pick, 2007). The surging motion of the airfoil can be controlled by the following equation:

$$l(t) = l_m \sin(k \cdot 2\pi ft + \varphi), \quad (3)$$

where $l(t)$ represents the instantaneous displacement of pitch–pivot–point in horizontal direction at time t , l_m is the corresponding surging amplitude. k is an adjustable parameter that is concerned with the relative frequency of surging motion. In the current study, five values of k are considered, i.e., $k = 0, 1, 2, 3$ and 4 . Different k may produce different shape of motion trajectory. In particular, $k = 0$ means the traditional motion. In addition, φ is the phase difference between pitch and surge, which also can influence the motion trajectory. The schematic diagram of motion trajectory with different combination of k and φ is shown in Fig. 1.

In the study of a flapping airfoil, the non-dimensional frequency (reduced frequency, St) is favorable, which is defined as $St = fc/U_\infty$, where U_∞ is the free stream velocity of oncoming flow. Since the MAVs are inspired by small animals in nature, the Reynolds number based on the free stream velocity and the chord length is then selected as $Re = U_\infty c/\nu = 500$, where ν is the kinematic viscosity. For convenience, the details of parameters used in this work are summarized in Table 1.

2.2. Propulsion mode

The mean thrust coefficient $C_{t,mean}$ and mean power coefficient $C_{p,mean}$ can be viewed as the generated thrust and consumed power by airfoil, respectively. They are defined as following equations:

$$C_{t,mean} = \frac{-\frac{1}{T} \int_0^T F_x(t) dt}{0.5\rho c U_\infty^2}, \quad (4)$$

$$C_{p,mean} = \frac{-\frac{1}{T} \int_0^T p_{input} dt}{0.5\rho c U_\infty^3}. \quad (5)$$

And the consumed power and its coefficient can be expressed as follows:

$$p_{input} = F_y(t) \frac{dh(t)}{dt} + F_x(t) \frac{dl(t)}{dt} + M(t) \frac{d\theta(t)}{dt}, \quad (6)$$

$$C_p = C_{pl} - C_{pt} + C_{pm} = \frac{1}{U_\infty} \left[C_l \frac{dh(t)}{dt} - C_t \frac{dl(t)}{dt} + C_m \frac{d\theta(t)}{dt} c \right], \quad (7)$$

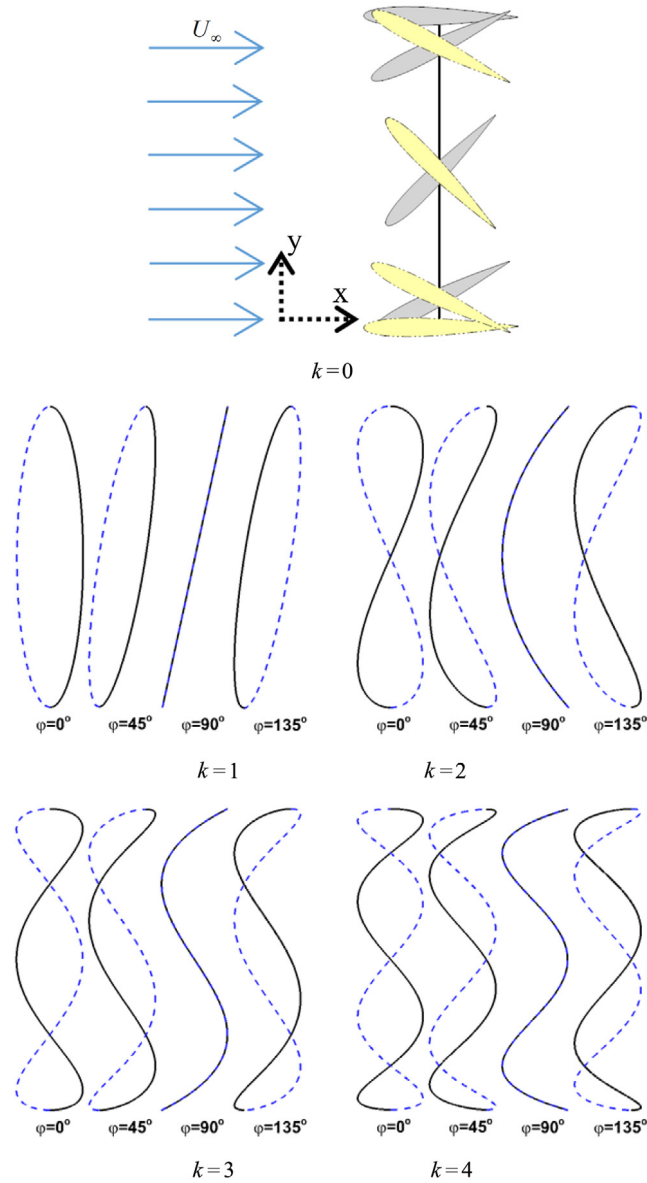


Fig. 1. Sketch of flow over a flapping foil with different motion trajectories. Solid line means the downstroke, dashed line means the upstroke.

where $F_x(t)$ and $F_y(t)$ are the instantaneous force in positive x direction and y direction at time t , respectively, and $M(t)$ is the generated torque about the pitching axis at time t . $C_l (= \frac{F_y(t)}{0.5\rho c U_\infty^2})$, $C_t (= \frac{-F_x(t)}{0.5\rho c U_\infty^2})$ and $C_m (= \frac{M(t)}{0.5\rho c^2 U_\infty^2})$ are the corresponding non-dimensional coefficients. p_{input} is the instantaneous consumed power, and T represents the period. Additionally, C_{pl} , C_{pt} and C_{pm} represent the power coefficients consumed by plunging motion, surging motion and pitching motion, respectively. Thus, the propulsive efficiency η_t can be defined as follows:

$$\eta_t = \frac{C_{t,mean}}{C_{p,mean}}. \quad (8)$$

The propulsive efficiency is considered as $\eta_t = 0$ when the mean force in x direction becomes drag during a flapping period.

2.3. Numerical method and validation

In this work, flows over a flapping foil are simulated by using an immersed boundary-lattice Boltzmann method (IB-LBM) (Wu and Shu, 2009). For two-dimensional viscous and incompressible flows, the governing equations in the framework of

IB-LBM can be written as

$$f_\alpha(\mathbf{x} + \mathbf{e}_\alpha \delta t, t + \delta t) - f_\alpha(\mathbf{x}, t) = -\frac{1}{\tau} (f_\alpha(\mathbf{x}, t) - f_\alpha^{eq}(\mathbf{x}, t)) + F_\alpha \delta t, \quad (9)$$

$$F_\alpha = \left(1 - \frac{1}{2\tau}\right) \omega_\alpha \left(\frac{\mathbf{e}_\alpha - \mathbf{u}}{c_s^2} + \frac{\mathbf{e}_\alpha \cdot \mathbf{u}}{c_s^4} \mathbf{e}_\alpha \right) \cdot \mathbf{f}, \quad (10)$$

$$\rho \mathbf{u} = \sum_\alpha \mathbf{e}_\alpha f_\alpha + \frac{1}{2} \mathbf{f} \delta t. \quad (11)$$

where f_α is the distribution function, and f_α^{eq} is its corresponding equilibrium state. τ is the single relaxation time, δt is the time step, $\mathbf{e}_\alpha$ is the lattice velocity, ω_α are the coefficients related to the lattice velocity model, and $\mathbf{f}$ is the force density, which is determined by the boundary force density. The IB-LBM adopted in this work has been well validated and applied for simulation of flapping foil problems. For more details, one can refer to our previous work (Wu et al., 2015a, 2015b).

In the current study, the size of computational domain is $40c \times 30c$. Initially, the flow is stationary. Free stream velocity is imposed at the inlet (left side) and natural boundary condition is applied at the upper and lower sides and the outlet (right side) as well. A non-uniform mesh is used, which is fine and uniform around the foil and tail. The mesh spacing of uniform mesh is Δ . In the numerical simulations, twelve flapping periods are used. It should be noted that the flow becomes periodic after five flapping cycles. Same as the previous work (Wu et al., 2015a, 2015b), $\Delta = 0.00625c$, which is fine enough to achieve accurate results, is used in the following simulations.

3. Results and discussion

After the brief introduction of numerical method employed, the influence of motion trajectory on the aerodynamic performance of a flapping foil is studied in detail. The factors of interest in this work are the pitching amplitude (θ_m), the location of pitching axis (x_p), the surging amplitude (l_m), the phase difference between pitch and surge (φ) as well as the relative frequency of surging motion (k). The details of parameters used can be found in Table 1.

3.1. Effects of pitching amplitude and pitching axis location

Before introducing the surging motion into the traditional motion, the effects of pitching amplitude and pitching axis location are investigated firstly in this section. Figs. 2–4 present the variation of mean thrust coefficient $C_{t,mean}$, mean power coefficients consumed by plunging motion $C_{pl,mean}$ and pitching motion $C_{pm,mean}$ as well as propulsive efficiency η_t with respect to the reduced frequency, respectively. The purpose of this section is to find out the optimal combination of pitching amplitude and pitching axis location, which corresponds to the maximum values of mean thrust coefficient and propulsive efficiency.

From Fig. 2(a), it can be seen that $C_{t,mean}$ generally increases with St when $\theta_m < 60^\circ$. In particular, at the same St , it is found that $\theta_m = 30^\circ$ can generate the maximum $C_{t,mean}$, and it is followed by $\theta_m = 15^\circ$. When $\theta_m \geq 60^\circ$, however, $C_{t,mean}$ becomes negative, which implies that the mean drag appears. It should be pointed out that when the plunging amplitude is different, the appearance of $C_{t,mean}$ may occur at different θ_m . It implies that the transition from the propulsion to drag may depend on h_m . This change is consistent with the natural phenomenon, in which birds flap their wings with middle pitching amplitude to generate the thrust. The explanation is that as the pitching amplitude increases, the maximum angle of attack of the flapping foil is larger. As a result, more drag force may be generated during the flapping motion.

From Fig. 2(b), it is found that $C_{t,mean}$ also increases with St smoothly. However, the influence of x_p on $C_{t,mean}$ is not obvious as compared with that of θ_m in Fig. 2(a). When St is small (< 0.26), $C_{t,mean}$ at different x_p is close to each other. When St becomes larger and larger, it can be seen that the performance at $x_p = c/4$ is the best, which means that the highest $C_{t,mean}$ is produced.

Based on the results in Fig. 2, it can be concluded that the pitching amplitude has great impact on the mean thrust coefficient, while the effect of pitch-pivot-point location is relatively weak. Therefore, the optimal combination of θ_m and x_p corresponding to high mean thrust can be selected as $\theta_m = 30^\circ$ and $x_p = c/4$. On the other hand, although higher $C_{t,mean}$ can be achieved at higher St , more power is also consumed. Thus, the behavior of propulsive efficiency varying with reduced frequency may show some difference compared with mean thrust coefficient.

From Fig. 3(a), it is shown that the absolute value of $C_{pl,mean}$ increases with St monotonically for all θ_m considered. A similar variation trend of $C_{pm,mean}$ can be found in Fig. 3(b). In the meanwhile, it is noted that the plunging motion consumes much more power than the pitching motion. Moreover, for a fixed reduced frequency, more plunging power and less pitching power are consumed at smaller pitching amplitude as plotted in Fig. 3(a) and 3(b), respectively. As a consequence, the variation behavior of η_t with respect to St shows some difference.

For the cases of $\theta_m = 15^\circ$ and 30° , as seen in Fig. 3(c), η_t decreases with St gradually. For the case of $\theta_m = 45^\circ$, however, η_t first increases with St , and it reaches the maximum value at $St = 0.28$. Thereafter, η_t starts to decrease. Since the mean drag force is generated when $\theta_m = 60^\circ$, η_t is set to be zero.

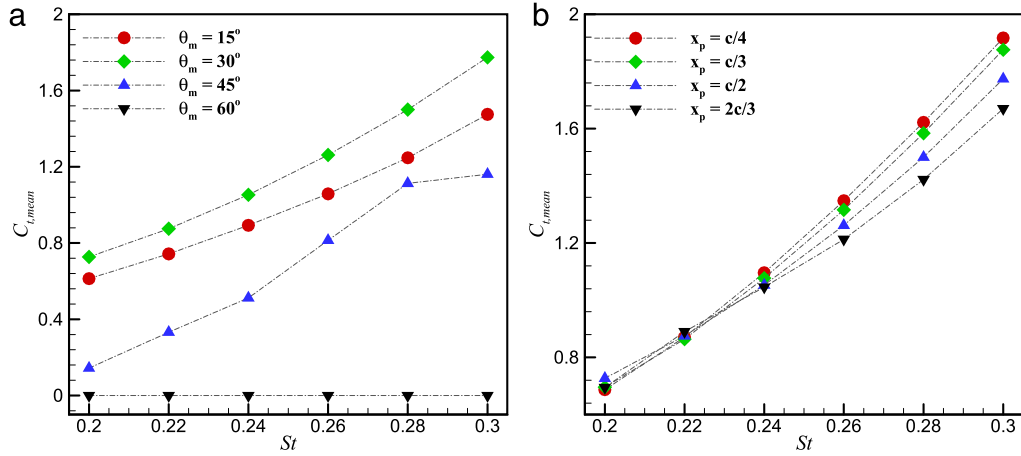


Fig. 2. Variation of mean thrust coefficient of a flapping foil with reduced frequency. (a) Different pitching amplitudes at $x_p = c/2$, (b) different pitching axis location at $\theta_m = 30^\circ$.

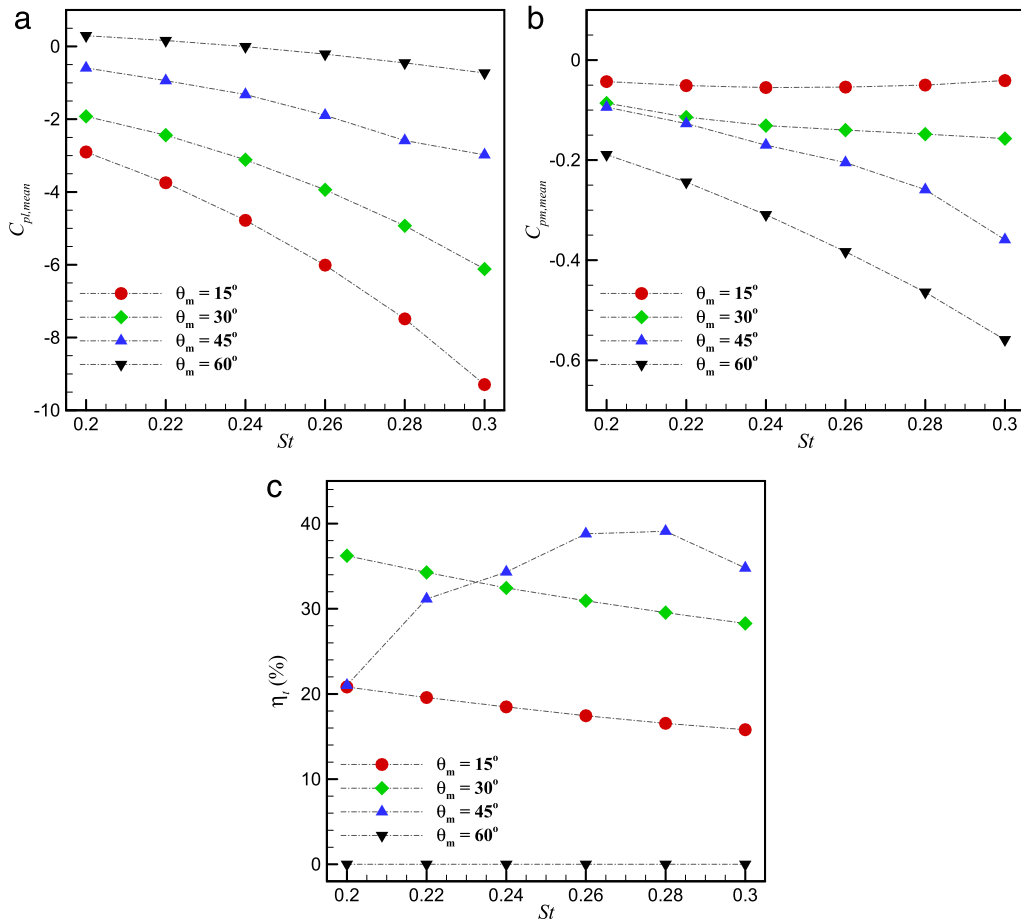


Fig. 3. Variation of (a) mean power coefficient consumed by plunging motion, (b) mean power coefficient consumed by pitching motion and (c) propulsive efficiency of a flapping foil with reduced frequency at $x_p = c/2$.

Same as the results in Fig. 3(a) and 3(b), it is found from Fig. 4(a) and 4(b) that the absolute values of both $C_{pl,mean}$ and $C_{pm,mean}$ increase with St for all x_p considered. In addition, the difference of $C_{pl,mean}$ at different x_p (Fig. 4(a)) is small for a given St , which implies that the pitch–pivot–point location only affects the plunging power slightly. For the case of $C_{pm,mean}$

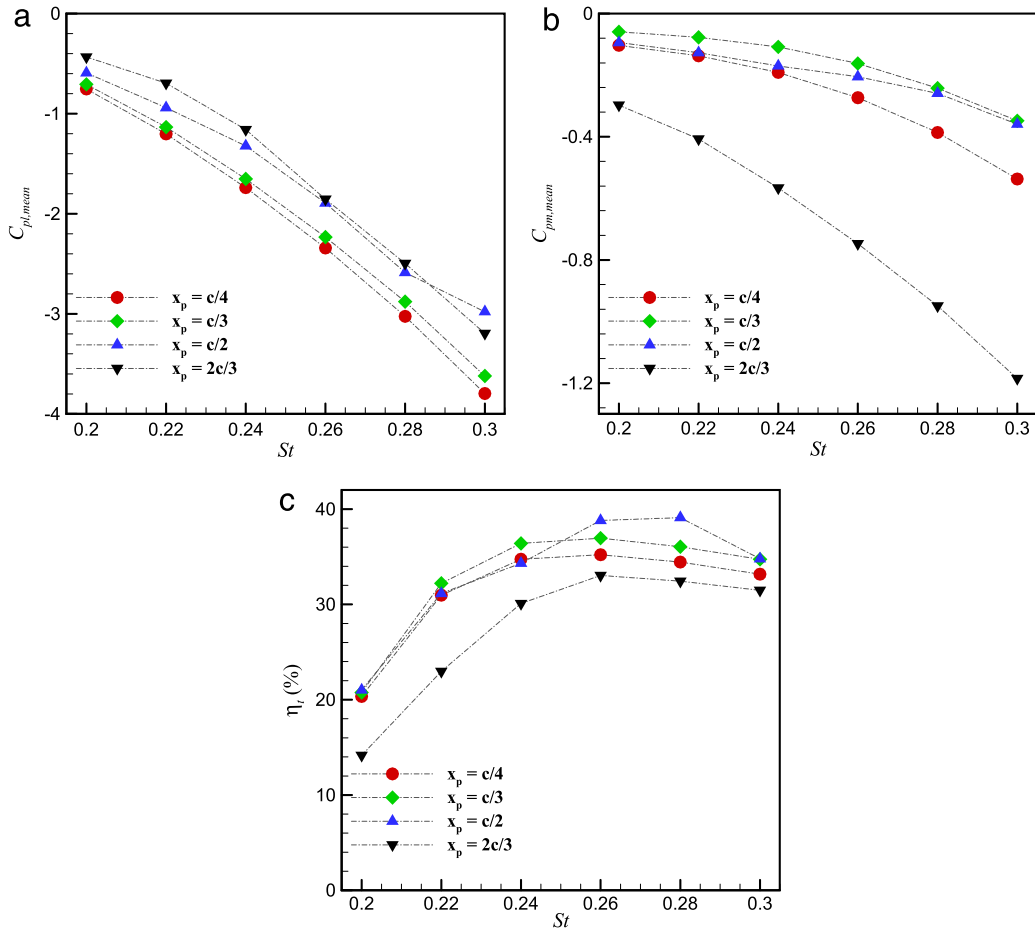


Fig. 4. Variation of (a) mean power coefficient consumed by plunging motion, (b) mean power coefficient consumed by pitching motion and (c) propulsive efficiency of a flapping foil with reduced frequency at $\theta_m = 45^\circ$.

(Fig. 4(b)), the similar situation can be observed when $x_p \leq c/2$. Nevertheless, the consumed pitching power is increased clearly when $x_p = 2c/3$. From Fig. 4(c), it then is known that with the increase of St , η_t first increases and then decreases smoothly. In particular, the maximum η_t can be achieved at $x_p = c/2$ and $St = 0.28$.

Based on the results in Figs. 3 and 4, it can be inferred that the propulsive efficiency is more influenced by the pitching amplitude than the pitching axis location. Thus, the optimal combination of pitching amplitude and pitching axis location, which produces the maximum propulsive efficiency ($\eta_t = 39.1\%$), can be chosen as $\theta_m = 45^\circ$ and $x_p = c/2$.

3.2. Effect of surging motion on mean thrust

It is known from Section 3.1 that high mean thrust appears under the condition of $\theta_m = 30^\circ$ and $x_p = c/4$ in the traditional motion of a flapping foil. To investigate the influence of surging motion on the mean thrust, such optimal parameter combination is selected. Moreover, to concentrate on the surging parameters, i.e., surging amplitude l_m , phase difference φ and relative surging frequency k , the reduced frequency is fixed at $St = 0.28$.

Fig. 5 plots the change of $C_{t,mean}$ with l_m and φ . To make comparison, the result of no surge motion is also included in the figure. In general, it can be known from the figure that the added surging motion influences $C_{t,mean}$ significantly. As shown in Fig. 5(a), when $l_m (=0.05c)$ is small, the different motion trajectory (different k) only slightly changes $C_{t,mean}$. With the increase of l_m , the influence of k becomes more and more obvious. In particular, $C_{t,mean}$ increases with l_m at $k = 1$ or 2 , which makes $C_{t,mean}$ be larger than that of no surge motion. At $k = 3$ or 4 , however, $C_{t,mean}$ first increases with l_m and then decreases. On the other hand, it also can be found that $C_{t,mean}$ with $\varphi = 180^\circ$ is always close to that with $\varphi = 0^\circ$ at $k = 1, 2$ or 3 . But at $k = 4$, the difference of $C_{t,mean}$ between $\varphi = 180^\circ$ and $\varphi = 0^\circ$ becomes larger and larger with the increase of l_m .

As can be seen from Fig. 5(b), at a fixed $l_m (=0.2c)$, the variation behavior of $C_{t,mean}$ with φ shows some difference at different k . When the relative surging frequency is small ($k = 1, 2$ and 3), the modification of $C_{t,mean}$ by phase difference is not obvious. But when the relative surging frequency is high ($k = 4$), $C_{t,mean}$ changes clearly with the increase of φ . Particularly,

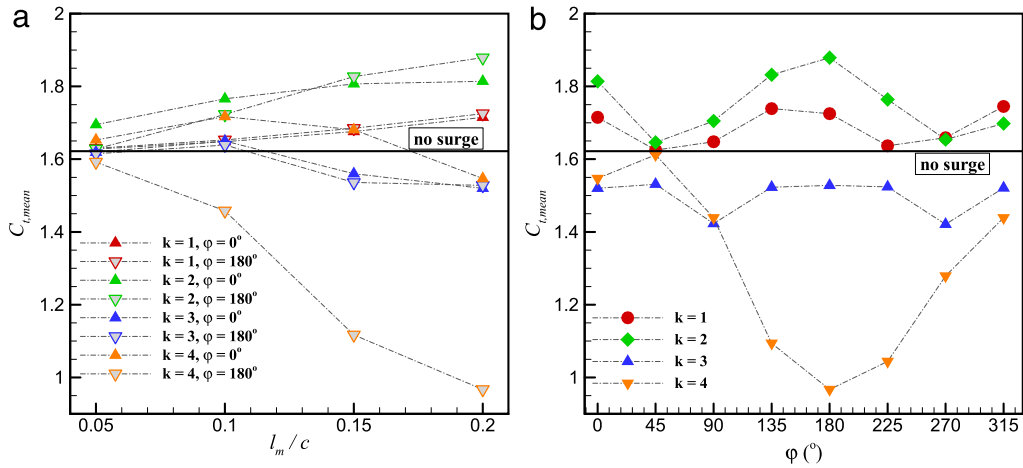


Fig. 5. Variation of mean thrust coefficient of a flapping foil with (a) surging amplitude and (b) phase difference with $l_m = 0.2c$ at $\theta_m = 30^\circ$, $x_p = c/4$ and $St = 0.28$.

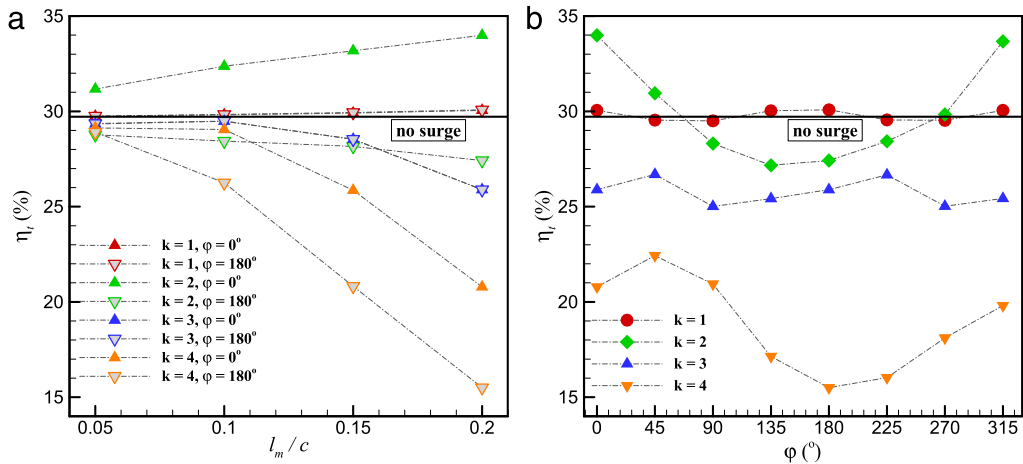


Fig. 6. Variation of propulsive efficiency of a flapping foil with (a) surging amplitude and (b) phase difference with $l_m = 0.2c$ at $\theta_m = 30^\circ$, $x_p = c/4$ and $St = 0.28$.

it can be known from Fig. 5 that the maximum $C_{t,mean}$ occurs at $k = 2$, $\phi = 180^\circ$ and $l_m = 0.2c$ with the introduction of surging motion. As compared with the traditional motion, it is increased from 1.622 to 1.879 (improvement is 15.8%).

Besides the mean thrust, it is also necessary to check the corresponding propulsive efficiency under the condition of $\theta_m = 30^\circ$, $x_p = c/4$ and $St = 0.28$. Fig. 6 provides the variation of η_t with l_m and ϕ . Again, the result of no surge motion is involved. From Fig. 6(a), it is found that η_t at $k = 1$ is close to that of traditional motion for all l_m considered. For the cases of $k = 3$ and 4, η_t is always smaller than that of traditional motion. The same behavior can be observed at $k = 2$ and $\phi = 180^\circ$. On the contrary, η_t at $k = 2$ and $\phi = 0^\circ$ is improved clearly compared with the traditional motion.

From Fig. 6(b), it is known that η_t at $k = 1$ is not sensitive to ϕ for a given fixed $l_m (=0.2c)$. A similar variation trend is found at $k = 3$. For the cases of $k = 2$ and 4, however, it is noted that η_t varies with ϕ significantly. Moreover, η_t can be enhanced at some specific values of ϕ . In particular, the maximum η_t can be achieved at $\phi = 0^\circ$. Compared with the traditional motion, it is increased from 29.7% to 34% (improvement by a factor of 1.145). Correspondingly, $C_{t,mean}$ is increased from 1.622 to 1.814 (improvement is 11.8%).

In the traditional motion, due to the symmetric motion trajectory, the mean lift coefficient $C_{l,mean}$ is zero. When the surging motion is added, the resultant trajectory may be no longer top-bottom symmetric under some specific conditions (shown in Fig. 1). Thus, the generated $C_{l,mean}$ is non-zero, which is presented in Fig. 7. Since the trajectory is still top-bottom symmetric at $k = 2$ or 4, $C_{l,mean}$ in Fig. 7 is also zero under the condition of $\theta_m = 30^\circ$, $x_p = c/4$ and $St = 0.28$, which is the same as the situation of traditional motion. Nevertheless, it should be indicated that when the reduced frequency further increases, the wake of flapping foil may become asymmetric at $k = 2$ or 4, and then $C_{l,mean}$ may become non-zero. This is out of the scope of current study, and will not be discussed in detail.

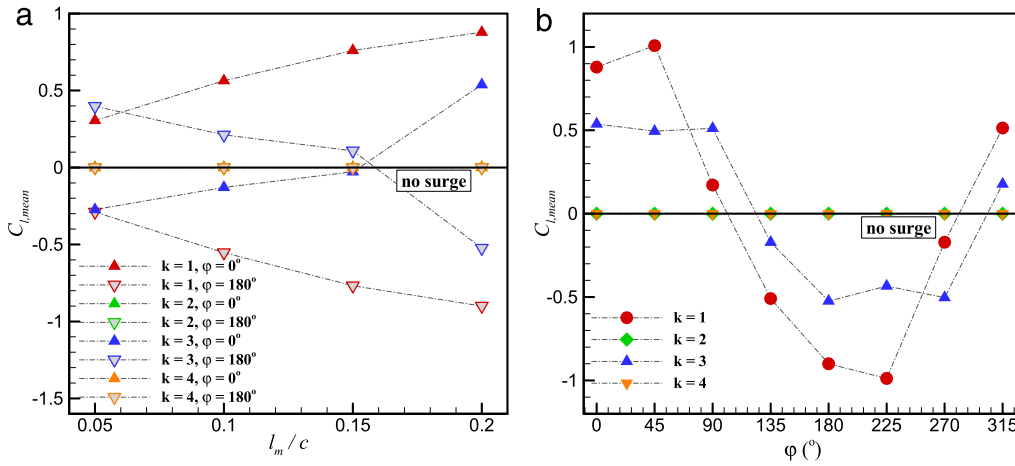


Fig. 7. Variation of mean lift coefficient of a flapping foil with (a) surging amplitude and (b) phase difference with $l_m = 0.2c$ at $\theta_m = 30^\circ$, $x_p = c/4$ and $St = 0.28$.

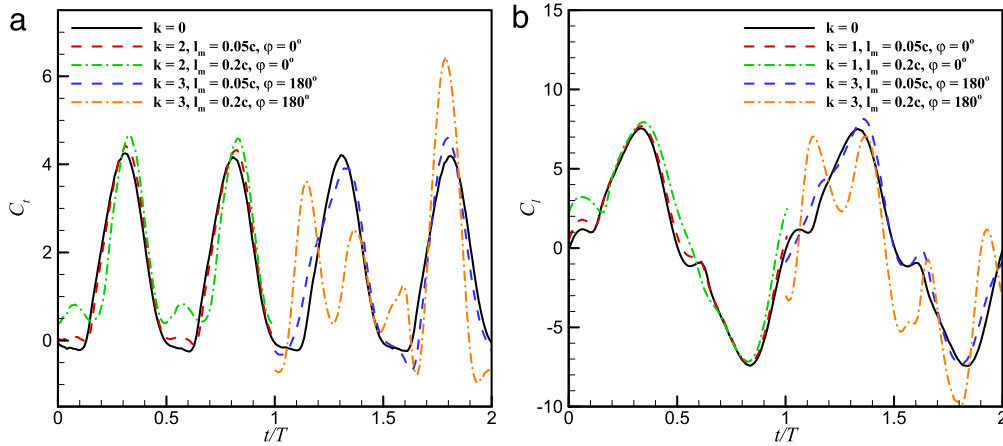


Fig. 8. Evolution of (a) thrust coefficient and (b) lift coefficient of a flapping foil during a flapping period at $\theta_m = 30^\circ$, $x_p = c/4$ and $St = 0.28$.

When the motion trajectory is not top–bottom symmetric at $k = 1$ or 3 , on the other hand, it can be found from Fig. 7 that either positive or negative $C_{l,mean}$ appears at different l_m and ϕ . With the increase of l_m , as shown in Fig. 7(a), $C_{l,mean}$ at $k = 1$ is always larger than zero at $\phi = 0^\circ$, and it is always less than zero at $\phi = 180^\circ$. However, for the case of $k = 3$, $C_{l,mean}$ at $\phi = 0^\circ$ increases from negative value to positive value, and the opposite trend is found at $\phi = 180^\circ$. With the increase of ϕ , as shown in Fig. 7(b), the variation trend of $C_{l,mean}$ at $k = 1$ and 3 is similar. In particular, it is known from Fig. 7 that the maximum $C_{l,mean}$ occurs at $k = 1$, $\phi = 45^\circ$ and $l_m = 0.2c$.

To further investigate effect of surging motion on thrust and lift forces, the evolution of thrust and lift coefficients during a flapping period T are displayed in Fig. 8. As seen from Fig. 8(a), the positive peak of C_t (around at $t/T = 3/10$ and $8/10$) is increased by the surging motion with $k = 2$ and $\phi = 0^\circ$. Moreover, a higher l_m can lead to a higher peak value. For the situation of $k = 3$, $l_m = 0.05c$ and $\phi = 180^\circ$, two positive peaks of C_t are decreased and increased, respectively. However, the negative peak of C_t (around at $t/T = 6/10$) is decreased. When l_m is increased up to $0.2c$, there are multiple peaks of C_t , and C_t at some instants is greatly reduced.

From Fig. 8(b), it is found that the surging motion with $k = 1$, $l_m = 0.05c$ and $\phi = 0^\circ$ can enhance the second positive peak of C_l (around at $t/T = 1/10$). When the surging amplitude is increased up to $l_m = 0.2c$, the second positive peak of C_l is further enhanced. Moreover, C_l at any instant is also improved. For the situation of $k = 3$, $l_m = 0.05c$ and $\phi = 180^\circ$, the positive peak of C_l (around at $t/T = 3/10$) is clearly enhanced compared with that of no surge motion. As l_m becomes $0.2c$, however, the negative peak of C_l (around at $t/T = 8/10$) is much lower. In addition, C_l at some instants is also decreased. All the results in Fig. 8 explain well the performance of $C_{t,mean}$ and $C_{l,mean}$ in Figs. 5 and 7.

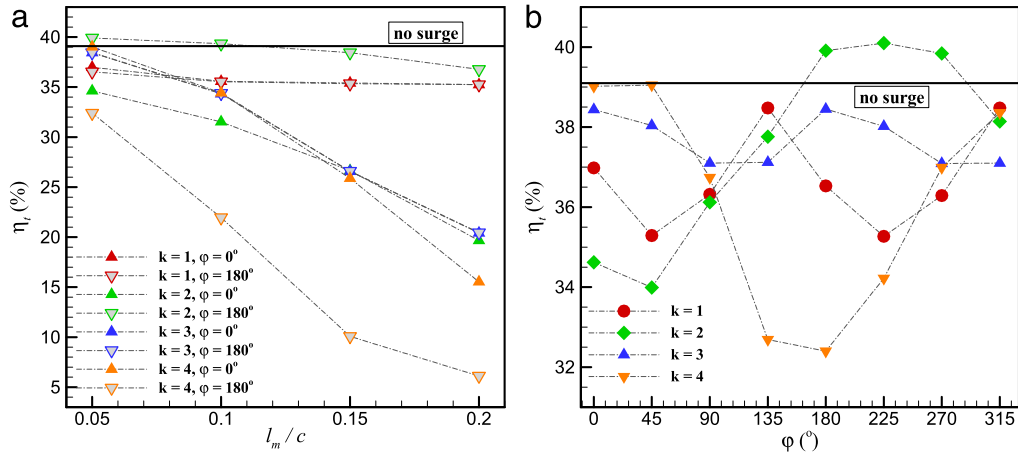


Fig. 9. Variation of propulsive efficiency of a flapping foil with (a) surging amplitude and (b) phase difference with $l_m = 0.05c$ at $\theta_m = 45^\circ$, $x_p = c/2$ and $St = 0.28$.

3.3. Effect of surging motion on propulsive efficiency

It is also known from Section 3.1 that the maximum propulsive efficiency for the traditional motion can be achieved under the condition of $\theta_m = 45^\circ$ and $x_p = c/2$. To check the effect of surging motion on the propulsive efficiency, such parameter combination is adopted. In addition, the reduced frequency is still fixed at $St = 0.28$.

Fig. 9 presents the variation of η_t with l_m and φ . Similarly, the result of no surge motion is provided again. As shown in Fig. 9(a), for the case of $k = 1$, η_t first decreases with l_m slightly, and then it nearly keeps constant as l_m further increases. However, when $k > 1$, it is found that η_t monotonically decreases with l_m . Moreover, the larger the relative surging frequency is, the faster the speed of decrease is. Compared with the traditional motion, the improvement of propulsive efficiency only occurs under the condition of $k = 2$ and $\varphi = 180^\circ$ when l_m is small ($\leq 0.1c$).

For a given $l_m (=0.05c)$ as plotted in Fig. 9(b), it can be observed that η_t at $k = 1$ or 3 changes with φ more smoothly than that at $k = 2$ or 4. Compared with the traditional motion, the phase difference reduces the propulsive efficiency under most situations. The efficiency enhancement only appears at $k = 2$ and $180^\circ \leq \varphi \leq 270^\circ$. In particular, the maximum η_t is obtained at $\varphi = 225^\circ$. Compared with the traditional motion, it is increased from 39.1% to 40.1% (improvement by a factor of 1.026).

Besides the propulsive efficiency, it is also needed to demonstrate the corresponding mean thrust under the same condition. Fig. 10 shows the variation of $C_{t,mean}$ with l_m and φ . In general, the variation trend of $C_{t,mean}$ in Fig. 10(a) is similar to that in Fig. 5(a), which means that higher $C_{t,mean}$ can be achieved at larger l_m . On the other hand, it is also noted that the difference of $C_{t,mean}$ at $k = 2$ or 4 between $\varphi = 0^\circ$ and 180° is always clear for all l_m considered, which is different from the results in Fig. 5(a).

At a small $l_m (=0.05c)$, as seen in Fig. 10(b), the variation trend of $C_{t,mean}$ is almost the same as η_t in Fig. 9(b). The only difference is that $C_{t,mean}$ can be improved by the surging motion under more situations. In particular, the maximum $C_{t,mean}$ in Fig. 10(b) just corresponds to the maximum η_t in Fig. 9(b). Compared with the traditional motion, it is increased from 1.113 to 1.464 (improvement is 31.5%).

Similar to Fig. 7, the mean lift coefficient also becomes non-zero due to the asymmetric motion trajectory at $k = 1$ or 3 under the condition of $\theta_m = 45^\circ$ and $x_p = c/2$. Thus, the change of $C_{l,mean}$ is illustrated in Fig. 11. Different from the results in Fig. 7, $C_{l,mean}$ at $k = 1$ and 3 is slightly modified by l_m , as shown in Fig. 11(a). Moreover, the absolute value of $C_{l,mean}$ at $k = 3$ is always larger than that at $k = 1$. As seen in Fig. 11(b), for a fixed $l_m (=0.05c)$, the maximum $C_{l,mean}$ at $k = 1$ and 3 appears at $\varphi = 225^\circ$ and 0° , respectively.

Similar to Fig. 8, the evolution of C_t and C_l during a flapping period T are provided in Fig. 12. From Fig. 12(a), it is found that the surging motion with $k = 2$, $l_m = 0.05c$ and $\varphi = 180^\circ$ can enhance the positive peak of C_t (around at $t/T = 3/10$ and $8/10$). Moreover, C_t during some time intervals is also larger than that of the traditional motion. When l_m is increased up to $0.2c$, the enhancement of C_t is much more. For the case of $k = 2$, $l_m = 0.05c$ and $\varphi = 0^\circ$, two positive peaks of C_t are clearly increased and decreased, respectively. Meanwhile, the increment is larger than the decrement. As l_m becomes $0.2c$, both positive and negative peaks of C_t with high magnitude can be observed, and C_t during time intervals is greatly reduced.

From Fig. 12(b), it is known that the surging motion with $k = 1$, $l_m = 0.05c$ and $\varphi = 180^\circ$ can improve both the positive peak of C_l (around at $t/T = 3/10$) and the second negative peak of C_l (around at $t/T = 5/10$). With the increase of l_m , it is found that the curve of C_l only changes a little. For the case of $k = 3$, $l_m = 0.05c$ and $\varphi = 0^\circ$, two positive peaks of C_l (around at $t/T = 0$ and $3/10$) are increased significantly. When l_m becomes $0.15c$, there is further increment of peaks of C_l . Moreover,

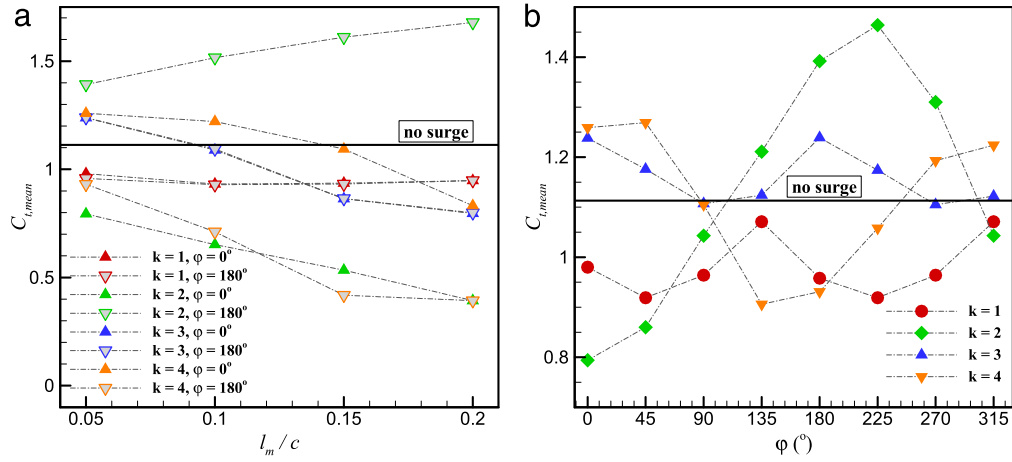


Fig. 10. Variation of mean thrust coefficient of a flapping foil with (a) surging amplitude and (b) phase difference with $l_m = 0.05c$ at $\theta_m = 45^\circ$, $x_p = c/2$ and $St = 0.28$.

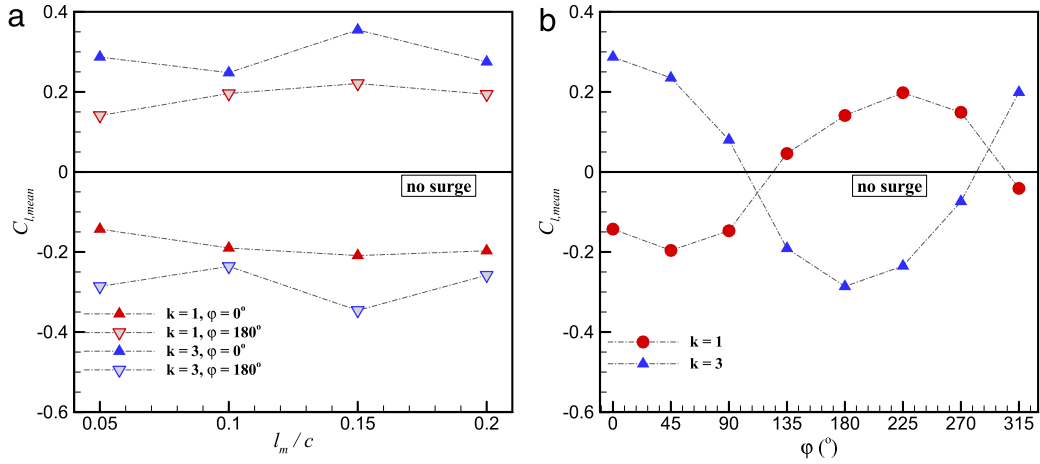


Fig. 11. Variation of mean lift coefficient of a flapping foil with (a) surging amplitude and (b) phase difference with $l_m = 0.05c$ at $\theta_m = 45^\circ$, $x_p = c/2$ and $St = 0.28$.

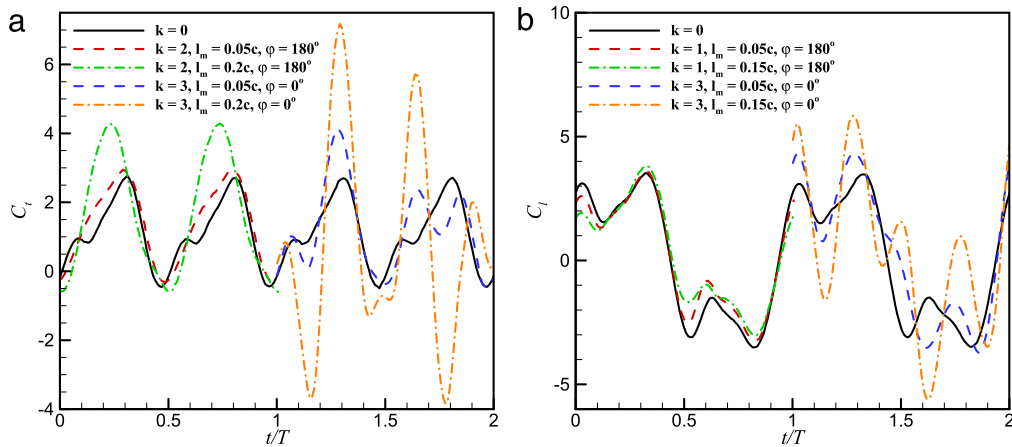


Fig. 12. Evolution of (a) thrust coefficient and (b) lift coefficient of a flapping foil during a flapping period at $\theta_m = 45^\circ$, $x_p = c/2$ and $St = 0.28$.

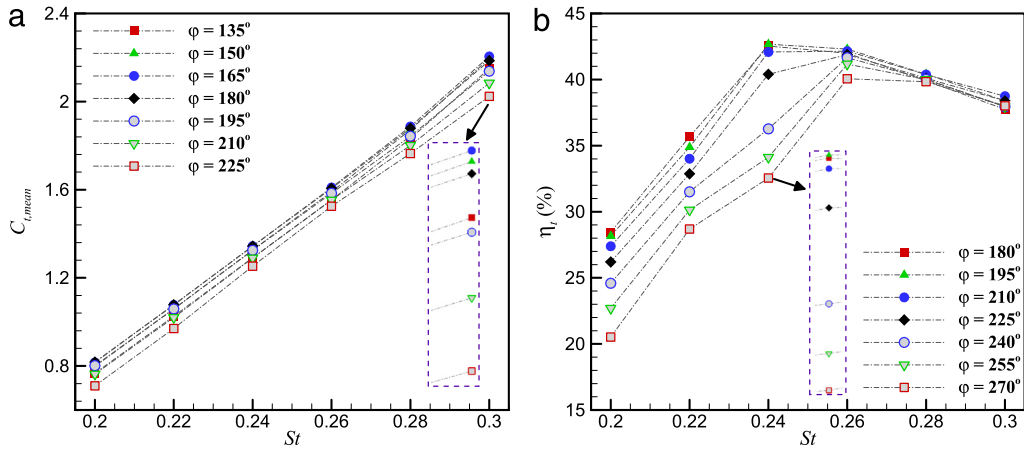


Fig. 13. Variation of (a) mean thrust coefficient at $\theta_m = 30^\circ$, $x_p = c/4$, $l_m = 0.2c$ and (b) propulsive efficiency at $\theta_m = 45^\circ$, $x_p = c/2$, $l_m = 0.05c$ with reduced frequency.

C_l during some time intervals is also increased more or less. Again, all the results in Fig. 12 explain well the behavior of $C_{t,mean}$ and $C_{l,mean}$ in Figs. 10 and 11.

3.4. Further exploration on mean thrust and propulsive efficiency

From the above analysis, it can be seen that the added surging motion can obviously improve the mean thrust when using the optimal combination of θ_m and x_p (parameter combination 1) related to the maximum mean thrust of traditional motion. In the meanwhile, a high mean lift can be produced under some specific condition. On the other hand, the surging motion only slightly enhances the propulsive efficiency when using the optimal combination of θ_m and x_p (parameter combination 2) corresponding to the maximum propulsive efficiency of no surge motion. Moreover, the generated non-zero mean lift is also smaller than the situation of parameter combination 1. Since the reduced frequency is fixed at $St = 0.28$ in the above studies, the effect of surging motion on the mean thrust and propulsive efficiency are then further explored in this section by changing St . The relative surging frequency is fixed at $k = 2$ due to the high performance of mean thrust and propulsive efficiency. Fig. 13 presents $C_{t,mean}$ and η_t varying with St under the condition of parameter combination 1 and 2, respectively. Compared with the traditional motion, it can be seen that the participation of surging motion can further enhances $C_{t,mean}$ ($=2.205$ at $\varphi = 165^\circ$ and $St = 0.3$, improvement is 15.0%) and η_t ($=42.7\%$ at $\varphi = 195^\circ$ and $St = 0.24$, improvement by a factor of 1.092).

3.5. Modified plunging amplitude of traditional motion

It can be said that adding surging motion to the stroke is equivalent to increasing the velocity of flapping foil without surging motion. Then the traditional motion, which has the plunging amplitude leading to the velocity matching that of added surging motion, is also checked. Firstly, the maximum magnitude of velocity at different relative surging frequency k and phase difference φ under the condition of $\theta_m = 30^\circ$, $x_p = c/4$, $l_m = 0.2c$ and $St = 0.28$ is calculated, which is shown in Fig. 14. The results of traditional motion are also involved in the figure. Here, U1 represents the velocity magnitude corresponding to the traditional motion, and U2 to U5 represents the velocity magnitude corresponding to the surging motion at different φ . It is noted that some results of surging motion are the same at different φ , and then only the results with different maximum magnitudes are provided.

From Fig. 14, it is known that with the increase of k , the ratio of U2 (and U3, U4 and U5) to U1 becomes larger and larger. At a fixed k , there is a maximum ratio, i.e., 1.02 ($k = 1$, $\varphi = 90^\circ$), 1.077 ($k = 2$, $\varphi = 0^\circ$), 1.166 ($k = 3$, $\varphi = 90^\circ$) and 1.281 ($k = 4$, $\varphi = 0^\circ$). Then the plunging amplitude of traditional motion can be modified to match the velocity magnitude of surging motion. The calculated mean thrust coefficient $C_{t,mean}$ and propulsive efficiency η_t are compared with that of corresponding surging motion, which are listed in Table 2. Motion 1 means the traditional motion with modified plunging amplitude, and motion 2 means the corresponding surging motion.

From Table 2, it is found that the traditional motion with modified plunging amplitude can increase the mean thrust coefficient and propulsive efficiency at high relative surging frequency ($k = 3$ and 4). When the relative surging frequency is low ($k = 1$ and 2), the surging motion still performs better. This is consistent with the result in Sections 3.2 and 3.3 (surging motion at $k = 2$ can enhance both the mean thrust force and propulsive efficiency).

Alternatively, adding surging motion is also equivalent to directly increasing the plunging amplitude of traditional motion. Here, the quarter of path length of the surging motion is set as the modified plunging amplitude. As can be seen

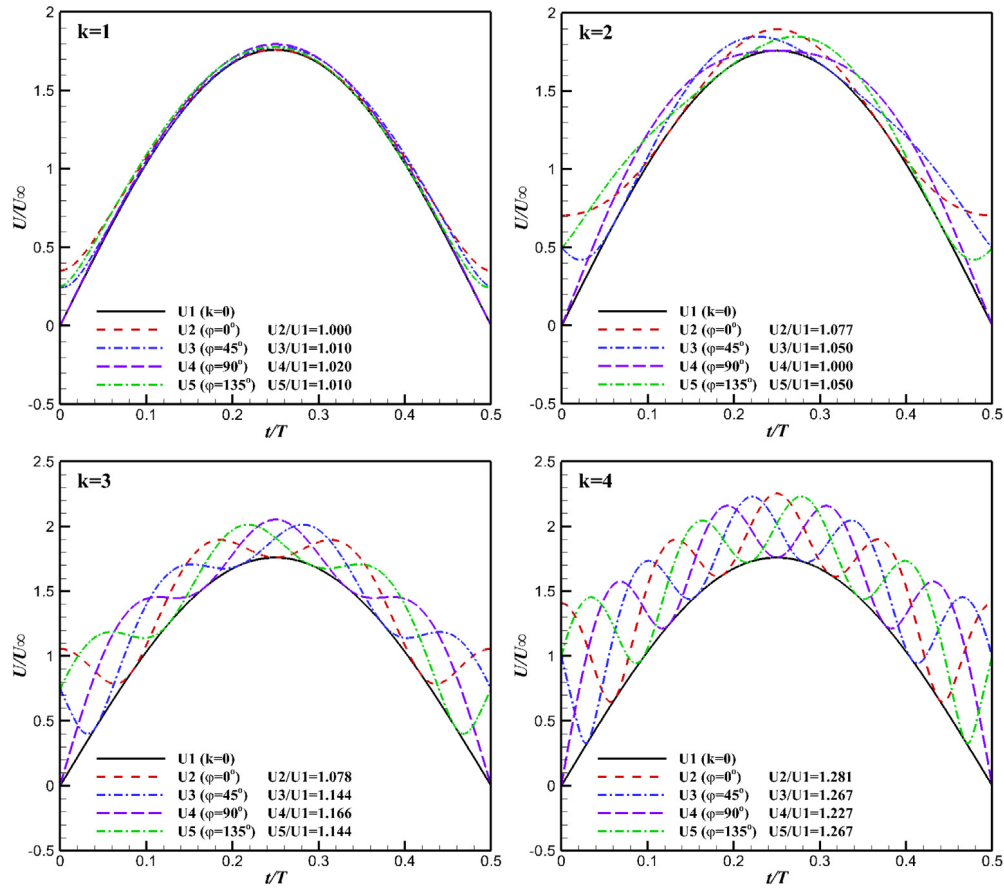


Fig. 14. Comparison of magnitude of velocity corresponding to traditional motion and surging motion at $\theta_m = 30^\circ$, $x_p = c/4$, $l_m = 0.2c$ and $St = 0.28$.

Table 2
Comparison of mean thrust coefficient and propulsive efficiency.

		$k = 1, \varphi = 90^\circ$	$k = 2, \varphi = 0^\circ$	$k = 3, \varphi = 90^\circ$	$k = 4, \varphi = 0^\circ$
$C_{t,mean}$	motion 1	1.701	1.91	2.225	2.712
	motion 2	1.975	2.229	1.702	1.814
η_t	motion 1	29.34%	28.1%	26.14%	24.03%
	motion 2	28.27%	32.57%	23.93%	19.43%

from Fig. 1 that with the increase of k , the path length becomes longer and longer. Then the calculated $C_{t,mean}$ and η_t are compared with that of corresponding surging motion, as listed in Table 3. The reduced frequency is still fixed at $St = 0.28$.

Based on the results in Table 3, it is shown that to increase the plunging amplitude of traditional motion can generally achieve better performance of $C_{t,mean}$ and η_t as compared with the surging motion, especially when relative surging frequency is high. This is consistent with the results in Table 2. The reason is that large k means large plunging amplitude in the tradition motion. However, it should be indicated that although $C_{t,mean}$ can be alternatively improved by increasing the plunging amplitude in the traditional motion, there might exist the limitation of plunging amplitude in some applications. Then, the way of introducing surging motion can be chosen to enhance $C_{t,mean}$. This might be the application of current study.

3.6. Mechanism of change of aerodynamic performance

To reveal the mechanism of change of aerodynamic performance due to the added surging motion, the force and torque behaviors should be analyzed carefully. Since both the mean thrust and propulsive efficiency are greatly modified by the added surging motion under the condition of $\theta_m = 30^\circ$ and $x_p = c/4$ (as presented in Section 3.2), such parameter combination together with $St = 0.28$ is selected. Other parameters are: $k = 2, 3$, $\varphi = 0^\circ, 180^\circ$ and $l_m = 0.2c$. As a reference, the traditional motion case with the same parameters is also considered.

Table 3

Comparison of mean thrust coefficient and propulsive efficiency.

		$k = 1$	$k = 2$	$k = 3$	$k = 4$
$C_{t,mean}$	motion 1	1.815	2.111	2.493	3.14
	motion 2	1.715	1.814	1.52	1.547
η_t	motion 1	28.69%	26.82%	24.78%	22.68%
	motion 2	30.37%	27.37%	25.87%	15.45%
$C_{t,mean}$	motion 3	1.149	1.213	1.286	1.364
	motion 4	0.958	1.392	1.239	0.931
η_t	motion 3	39.31%	39.55%	39.68%	39.68%
	motion 4	36.53%	39.91%	38.45%	32.4%

motion 1: traditional motion with modified plunging amplitude at $\theta_m = 30^\circ$ and $x_p = c/3$.

motion 2: surging motion at $\theta_m = 30^\circ$, $x_p = c/3$, $l_m = 0.2c$ and $\varphi = 180^\circ$.

motion 3: traditional motion with modified plunging amplitude at $\theta_m = 45^\circ$ and $x_p = c/2$.

motion 4: surging motion at $\theta_m = 45^\circ$, $x_p = c/2$, $l_m = 0.05c$ and $\varphi = 180^\circ$.

From Eq. (7), it is found that the plunging power coefficient C_{pl} is determined by the lift coefficient C_l and the plunging velocity $(dh/dt)/U_\infty$, the surging power coefficient C_{pt} is determined by the thrust coefficient C_t and the surging velocity $(dl/dt)/U_\infty$, and the pitching power coefficient C_{pm} is determined by the torque coefficient C_m and the angular velocity $(d\theta/dt)/U_\infty$. Thus, Fig. 15 provides the time evolution of C_t , $(dl/dt)/U_\infty$ (denoted as u_t), C_{pt} , C_l , $(dh/dt)/U_\infty$ (denoted as v_h), C_{pl} , C_m , $(d\theta/dt)/U_\infty$ (denoted as ω_θ) and C_{pm} over a flapping period.

As can be seen from Fig. 15(a), there is no negative value of C_t at $k = 2$ and $\varphi = 0^\circ$ during the whole flapping period. In the meanwhile, its peak value is also larger than that of traditional motion. As a result, higher $C_{t,mean}$ can be achieved. When φ becomes 180° , although the negative value of C_t appears at some instant, it is clearly increased during some time intervals. Then $C_{t,mean}$ is further enlarged. As k increases up to 3, although the positive peak C_t is larger, C_t is obviously decreased during some time intervals. This leads to a smaller $C_{t,mean}$. On the other hand, since C_t and u_t have a bad synchronization at $k = 2$ and $\varphi = 0^\circ$, C_{pt} then is negative during most parts of period as shown in Fig. 15(b). The opposite behavior is found for the case of $k = 2$ and 3 together with $\varphi = 180^\circ$.

From Fig. 15(c), it is known that the magnitude of both positive and negative peaks of C_l is increased at $k = 2$ and $\varphi = 0^\circ$. When φ turns to 180° , the variation of C_l becomes smoother. Due to the top–bottom symmetric motion trajectory, however, $C_{l,mean}$ is still zero, which is the same as that of traditional motion. For the case of $k = 3$ and $\varphi = 0^\circ$, the symmetry of C_l is broken, and then $C_{l,mean}$ is non-zero. As shown in Fig. 15(d), since C_l and v_h always have the opposite signs, C_{pl} has two negative peaks during the first and second half period, respectively. Additionally, due to the larger amplitudes of C_l and v_h compared with C_t and u_t , respectively, the peak value of C_{pl} is much larger than that of C_{pt} , which then produces higher $C_{pl,mean}$. It implies that the plunging motion consumes much more power than the surging motion. As shown in Fig. 15(e) and 15(f), on the other hand, since $C_{pm,mean}$ is very small, it is known that the influence of added surging motion on C_m and C_{pm} can be neglected.

To closely study the variation of C_t and C_l , it is necessary to check the flow field near the flapping foil. For qualitative demonstration, Fig. 16 illustrates the vorticity and pressure coefficient contours correspondingly. Since the variation behavior of C_t and C_l at $k = 2$ and $\varphi = 0^\circ$ is similar to that at $k = 2$ and $\varphi = 180^\circ$, only the results at $k = 2$ and $\varphi = 180^\circ$ are presented.

For the traditional motion as shown in Fig. 16(a), it is observed that a single vortex is shed alternately from the trailing edge of flapping foil and rotates in opposite direction. This is the famous reversed Karman vortex street, which can produce the thrust force with the explanation of momentum conservation law. When the foil reaches its medium position of plunging motion at $t/T = 1/4$ and $3/4$, the vortex street and pressure coefficient distribution are symmetric about the horizontal axis. A similar phenomenon can be found at $t/T = 2/4$ and $4/4$.

When the surging motion is introduced, the flow pattern becomes complicated. For the case of $k = 2$ and $\varphi = 180^\circ$ as shown in Fig. 16(b), it is found that more than one vortex is shed from the trailing edge. The reason may be that the surging motion obviously causes a pair of vortices (multiple vortices) generated around the leading edge which are developing along the foil surface at $t/T = 2/4$ and $4/4$, and they have departed from the surface of foil at $t/T = 1/4$ and $3/4$, respectively. On the other hand, it also can be known from Fig. 16(b) that although the surging motion can modify the pressure distribution, it is still symmetric about the horizontal axis at $t/T = 1/4$ and $3/4$ or at $t/T = 2/4$ and $4/4$. As the motion trajectory is top–bottom asymmetric at $k = 3$, as plotted in Fig. 16(c), the flow pattern of multiple vortices shedding still appears. However, no symmetry of vortex street and pressure coefficient distribution can be observed.

To further make a quantitative investigation, the comparison of local vorticity around the foil and the pressure coefficient distribution along the foil surface at some typical instants are provided in Figs. 17 and 18.

As shown in Fig. 17(a), three typical instants during the time history of thrust coefficient are denoted, i.e., $t/T = 5/32$ (s_1), $9/32$ (s_2) and $25/32$ (s_3). At instant s_1 , the foil rotates in the counterclockwise direction with a high angular velocity $\omega_\theta = 0.51$. As plotted in Fig. 17(b), a clockwise (negative) leading edge vortex (LEV) is developing along the upper surface

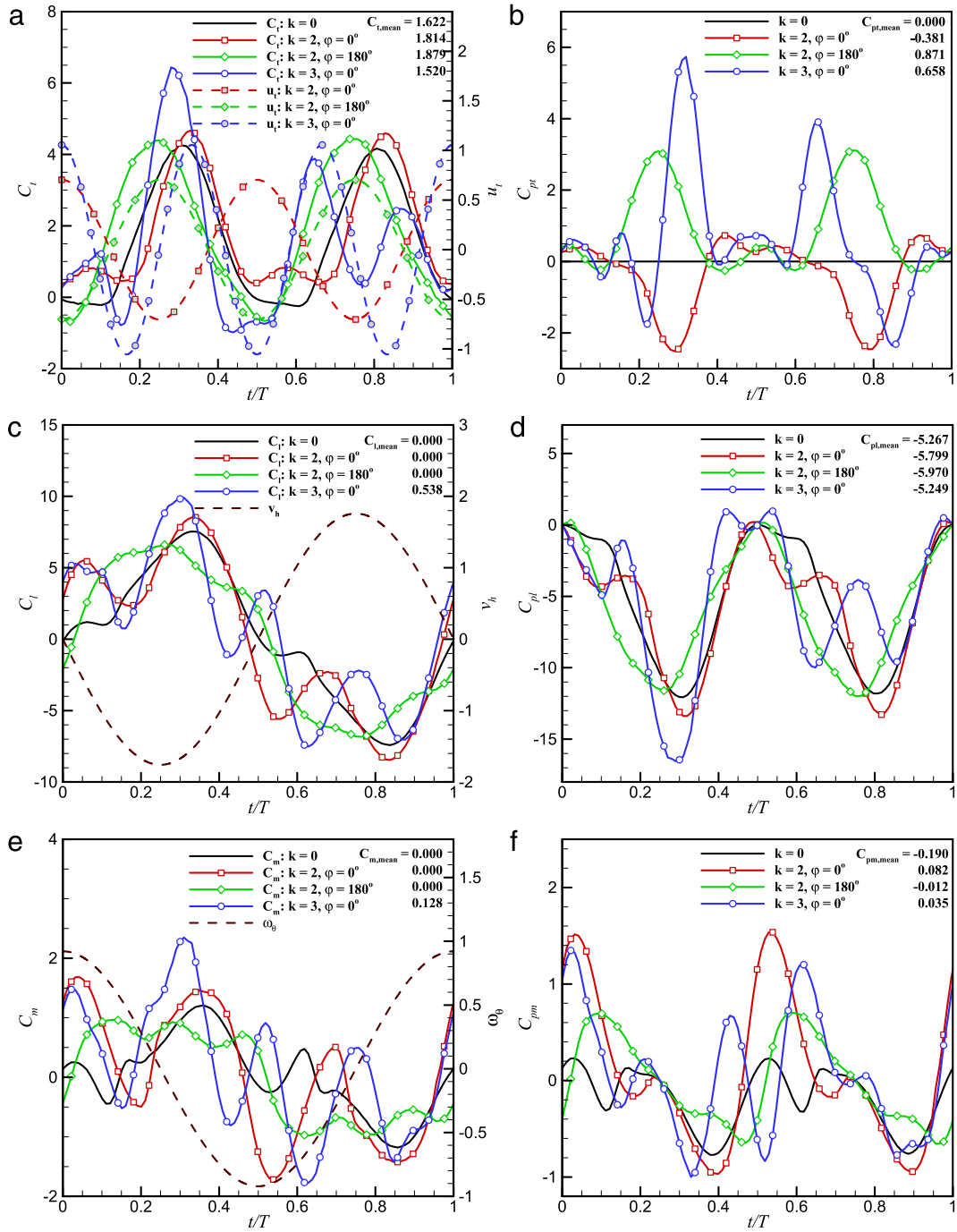


Fig. 15. Time evolution of (a) thrust coefficient and surging velocity, (b) power consumed by surging motion; (c) lift coefficient and heaving velocity, (d) power consumed by plunging motion; (e) torque coefficient and angular velocity, and (f) power consumed by pitching motion for a flapping foil at $\theta_m = 30^\circ$, $x_p = c/4$, $l_m = 0.2c$ and $St = 0.28$.

of the flapping foil. For the traditional motion ($C_t = 0.768$), a counterclockwise (positive) vortex with the same magnitude is also developing along the lower surface of the flapping foil. For the case of $k = 2$ and $\varphi = 180^\circ$ ($C_t = 3.099$), the foil has an additional positive horizontal velocity ($u_t = 0.27$). Then such positive lower surface vortex has been greatly compressed and is interacting with the trailing edge. In the meanwhile, the negative LEV slightly deviates from the upper surface. On the other hand, the pressure coefficient along the most part of lower surface is increased due to the positive horizontal velocity.

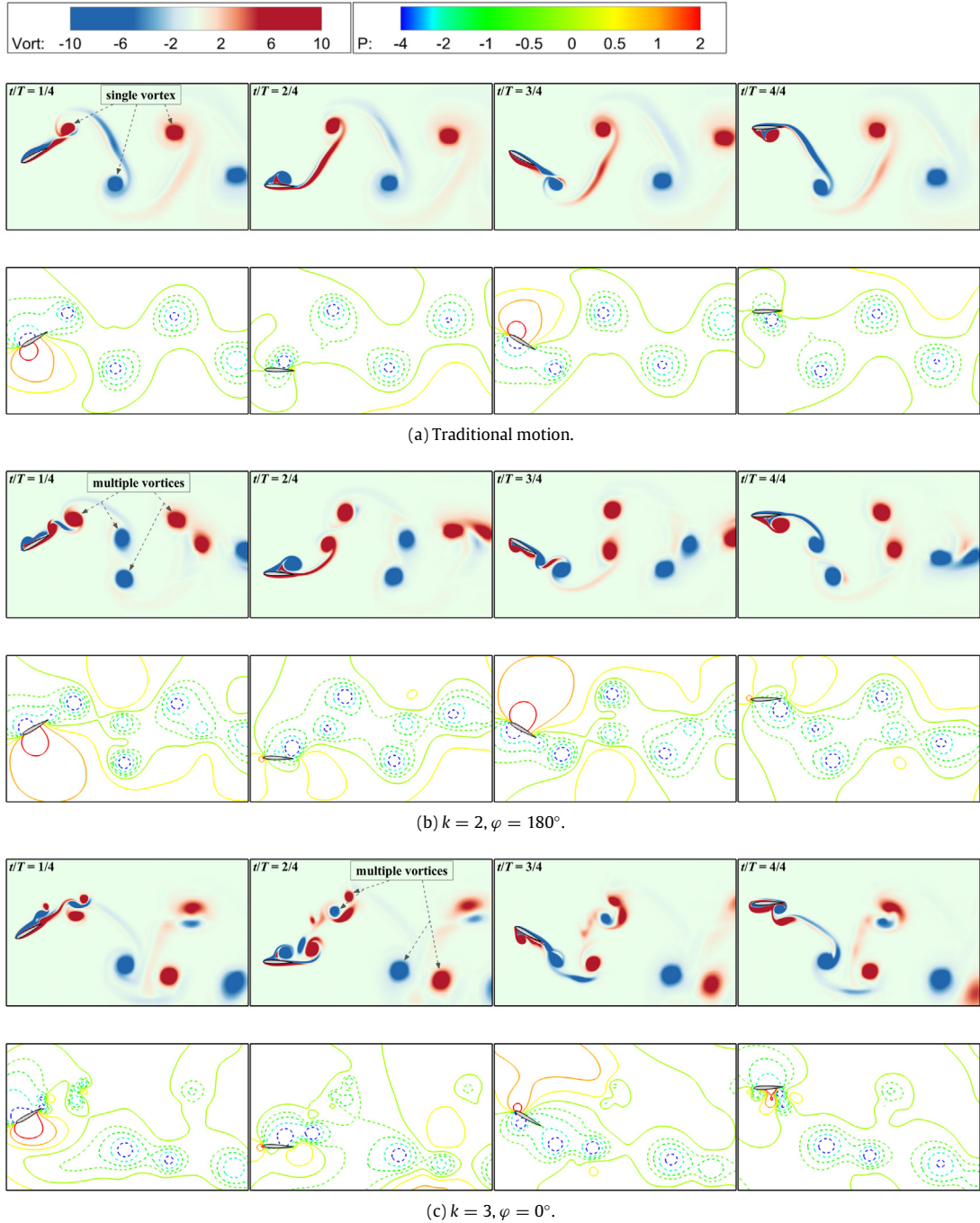
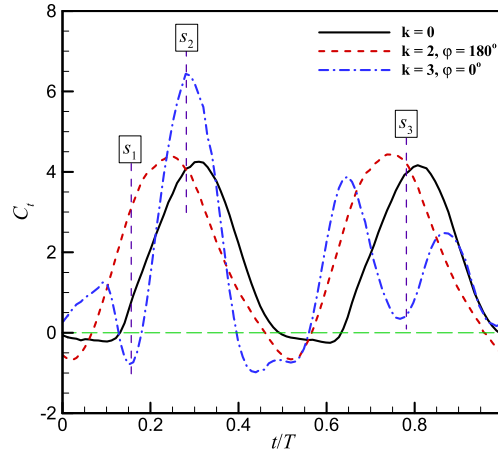
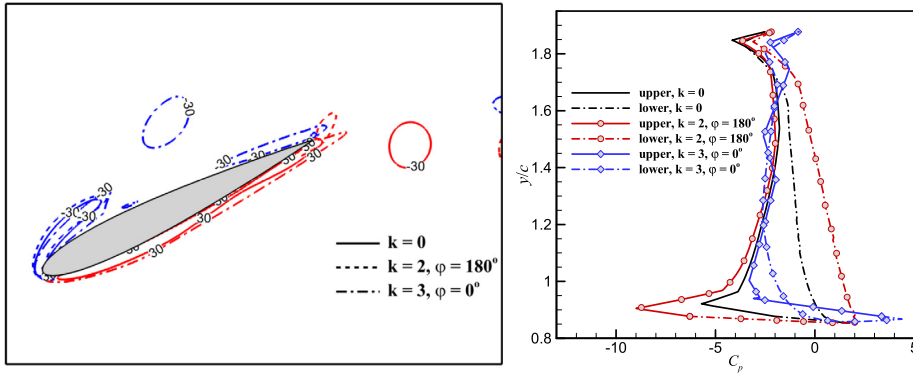
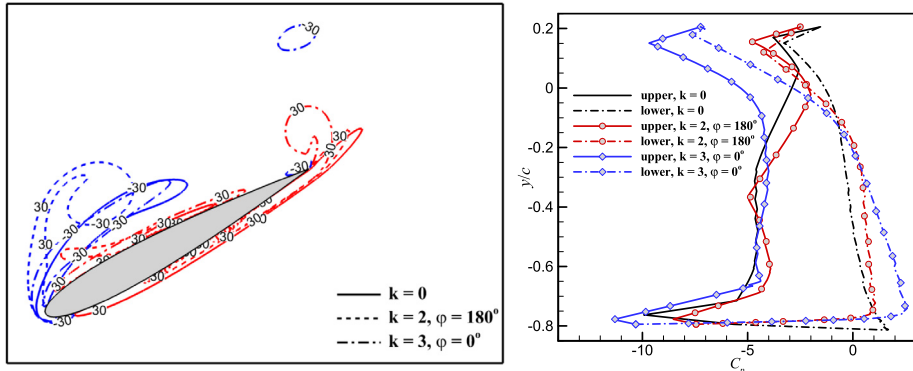


Fig. 16. Vorticity and pressure coefficient contours during a flapping period for flow over a flapping foil at $\theta_m = 30^\circ$, $x_p = c/4$, $l_m = 0.2c$ and $St = 0.28$.

Moreover, the pressure coefficient along the upper surface of leading edge is decreased. For the case of $k = 3$ and $\varphi = 0^\circ$ ($C_t = -0.758$), the horizontal velocity of foil is negative ($u_t = -1.04$). So the positive lower surface vortex is stretched, and the negative LEV is slightly compressed toward the upper surface. At the same time, the pressure coefficient along the lower surface is decreased significantly, whilst the pressure coefficient along the upper surface of leading edge is increased. Moreover, it is found that another negative vortex is interacting with the upper surface of trailing edge due to the surging motion. Such phenomenon may be called as the vortex capture. As a consequence, the pressure coefficient along the upper surface of trailing edge is increased more or less.



(a) Time history of thrust coefficient.

(b) Local vorticity comparison and pressure coefficient distribution at instant s_1 .(c) Local vorticity comparison and pressure coefficient distribution at instant s_2 .**Fig. 17.** Time history of thrust coefficient, local vorticity comparison and pressure coefficient distribution at three typical instants.

At instant s_2 , the foil rotates in the clockwise direction with a low angular velocity $\omega_\theta = -0.18$. As plotted in Fig. 17(c), the negative LEV becomes larger. For the traditional motion ($C_t = 4.08$), the positive lower surface vortex becomes longer. For the case of $k = 2$ and $\varphi = 180^\circ$ ($C_t = 4.051$), the horizontal velocity of foil becomes higher ($u_t = 0.65$). Then this positive vortex moves to the back part of lower surface, and the negative LEV is going to shed off due to the surging motion. In the meanwhile, the pressure coefficient along the front part of lower surface is increased, and there is also slight increase of pressure coefficient along the upper surface of leading edge. On the other hand, the pressure coefficient around the trailing edge is decreased. For the case of $k = 3$ and $\varphi = 0^\circ$ ($C_t = 6.433$), the horizontal velocity of foil becomes positive ($u_t = 0.59$). So the positive lower surface vortex has moved to the trailing edge and is going to shed off, whilst the negative LEV slightly

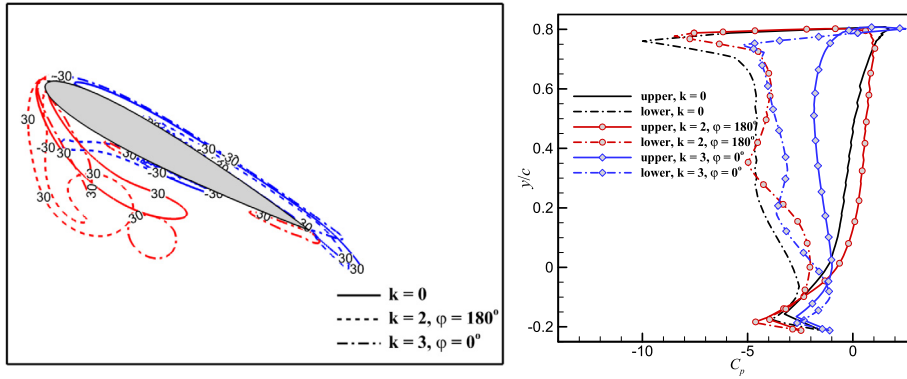
(d) Local vorticity comparison and pressure coefficient distribution at instant s_3 .

Fig. 17. (continued)

deviates from the upper surface. At the same moment, the pressure coefficient along the front part of lower surface is further increased, whilst the pressure coefficient along the upper surface of leading edge is decreased slightly. Moreover, the pressure coefficient around the trailing edge is decreased greatly.

After half of a flapping period and at instant s_3 , the angular velocity of foil becomes $\omega_\theta = 0.18$. As plotted in Fig. 17(d), the local vorticity and pressure coefficient distribution of traditional motion case ($C_l = 4.08$) and $k = 2$ and $\varphi = 180^\circ$ case ($C_l = 4.051$ and $u_t = 0.65$) are antisymmetric to that at instant s_2 . For the case of $k = 3$ and $\varphi = 0^\circ$ ($C_l = 0.444$), however, the horizontal velocity of foil becomes negative again ($u_t = -0.59$). Thus the negative vortex covers almost the whole part of upper surface, and the positive LEV along the lower surface is compressed. In the meanwhile, the pressure coefficient along the upper surface is decreased as compared with other two cases, whilst the pressure coefficient along the lower surface of leading edge is increased. Moreover, there is another positive vortex interacting with the lower surface of trailing edge. As a result, the pressure coefficient is increased correspondingly.

Similarly, two typical instants during the time history of lift coefficient are denoted in Fig. 18(a), i.e., $t/T = 10/32$ (s_4) and $24/32$ (s_5). Since instant s_4 ($\omega_\theta = -0.35$) just follows instant s_2 , as shown in Fig. 18(b), the local vorticity continues to develop at this instant. For the traditional motion ($C_l = 7.403$), the positive vortex along the lower surface is moving toward the trailing edge. For the case of $k = 2$ and $\varphi = 180^\circ$ ($C_l = 5.929$), the horizontal velocity of foil is reduced ($u_t = 0.5$). At this moment, such positive vortex is still located at the back part of lower surface, but the negative LEV has been broken into two parts. In the meanwhile, the lower surface pressure coefficient distribution is similar to that at instant s_2 , whilst the pressure coefficient along the upper surface is clearly increased. For the case of $k = 3$ and $\varphi = 0^\circ$ ($C_l = 9.841$), the horizontal velocity of foil is enhanced greatly ($u_t = 0.97$). Then the positive lower surface vortex continues to be stretched and is further away from the trailing edge, whilst the negative LEV is also stretched and deviates further from the upper surface. Meanwhile, the pressure coefficients distribution along both the lower surface and the upper surface are similar to that at instant s_2 .

Instant s_5 is followed by instant s_3 , and the angular velocity becomes zero at this instant. As shown in Fig. 18(c), the local vorticity develops toward that at instant s_3 . For the traditional motion ($C_l = -5.774$), the negative vortex along the upper surface is relatively shorter. For the case of $k = 2$ and $\varphi = 180^\circ$ ($C_l = -6.805$), the horizontal velocity of foil is highest ($u_t = 0.7$). Then this negative vortex tries to move toward the back part of upper surface, and the positive LEV along the lower surface has already been in the process of shedding. Meanwhile, the pressure coefficient along the front part of upper surface is increased, and the pressure coefficient along the lower surface of leading edge is also increased slightly. Nevertheless, the pressure coefficient around the trailing edge is still decreased. For the case of $k = 3$ and $\varphi = 0^\circ$ ($C_l = -2.256$), the horizontal velocity of foil also becomes zero. Then the negative upper surface vortex is nearly the same as that of traditional motion, but the positive LEV deviates from the lower surface. On the other hand, the pressure coefficient distribution is similar to that at instant s_3 . In particular, the vortex capture occurs at the back part of lower surface. Consequently, the corresponding pressure coefficient is increased greatly, which is even larger than that along the back part of upper surface.

The results in Figs. 15–18 and the corresponding analysis explain well the variation of mean thrust coefficient, mean lift coefficient and propulsive efficiency in Figs. 5–7.

4. Conclusions

In this study, the influence of motion trajectories on the aerodynamic characteristics of flapping airfoil is investigated through the numerical simulations. The motion trajectory includes a horizontal motion (surge) and a vertical motion (plunge) with different amplitudes and various oscillating frequencies. In addition, the airfoil executes pitching motion with the selected amplitude synchronously. To explore the mechanism of flapping airfoil, a NACA0012 airfoil is adopted with a low Reynolds number at $Re = 500$. After simulations of single flapping foil, it is identified that the combination of $\theta_m = 30^\circ$

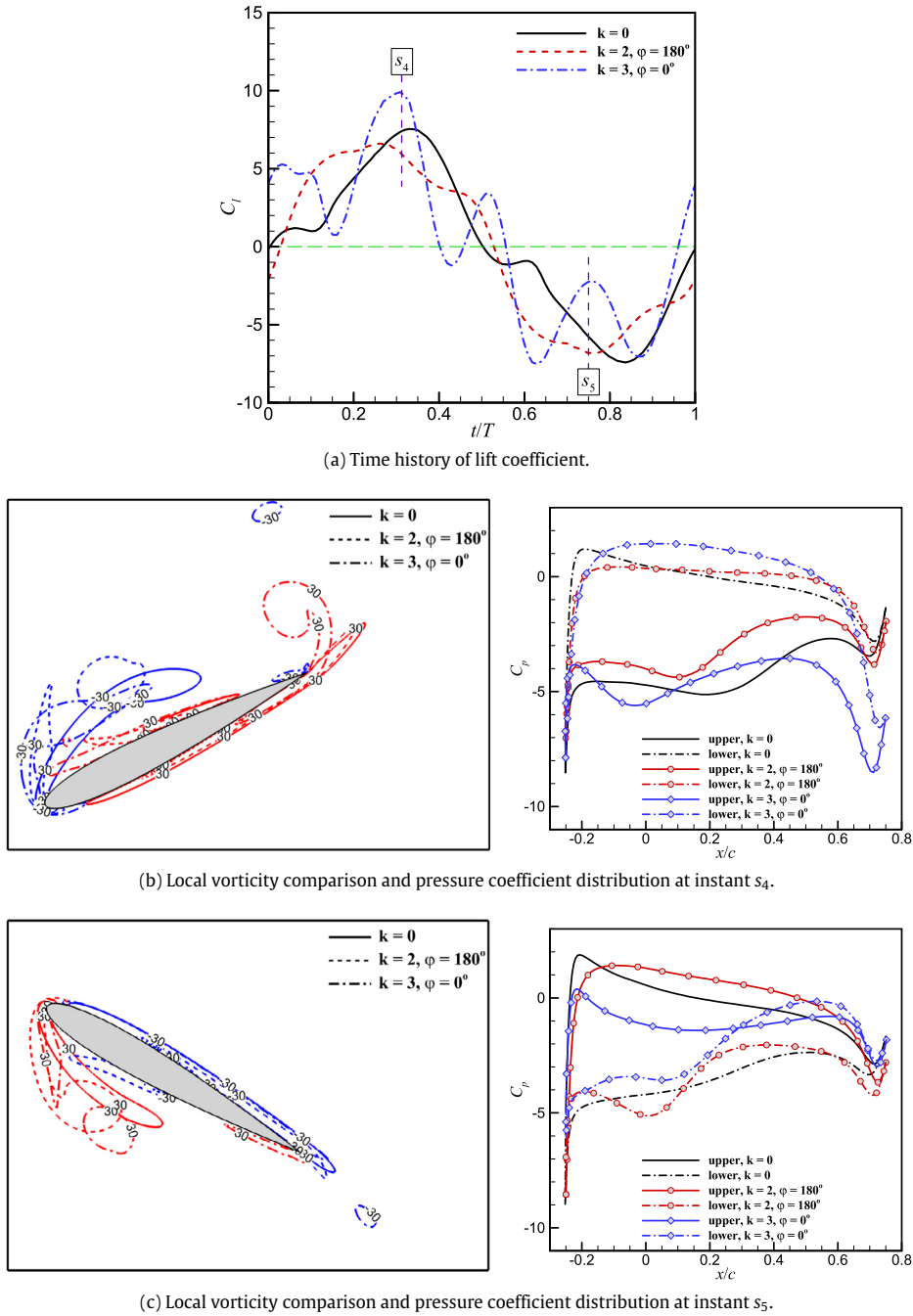


Fig. 18. Time history of lift coefficient, local vorticity comparison and pressure coefficient distribution at two typical instants.

and $x_p = c/4$ is suitable for achieving high mean thrust, and the combination of $\theta_m = 45^\circ$ and $x_p = c/2$ can be for high propulsive efficiency achievement. After that, the systematic simulations of flapping foil with added surging motion are performed. Based on the numerical results, some valuable conclusions can be obtained, and they are listed as follows:

1. The surging amplitude l_m and relative surging frequency k have a great role on the mean thrust force $C_{t,mean}$. To improve $C_{t,mean}$, high l_m with $k = 1$ or 2 should be selected. Moreover, the phase difference φ also can change $C_{t,mean}$, and the maximum $C_{t,mean}$ occurs at $\varphi = 180^\circ$ together with $k = 2$.

2. The added surging motion is less suitable for the enhancement of propulsive efficiency η_t . There is slight improvement of η_t only under the condition of small l_m with $k = 2$.
3. The added surging motion also can generate the non-zero mean lift force $C_{l,mean}$ at $k = 1$ or 3. Moreover, with the increase of l_m , $C_{l,mean}$ can be increased more or less.
4. The phenomenon of multiple vortices occurs when the surging motion is added. Additionally, it is revealed that the generation of multiple vortices and resultant vortex capture is one of the factors that are helpful to enhance the instantaneous thrust and lift forces. Other factor might be kinematic angle of attack, which is not examined in this work.

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References

- Amiralaei, M.R., Alighanbari, H., Hashemi, S.M., 2010. An investigation into the effects of unsteady parameters on the aerodynamics of a low Reynolds number pitching airfoil. *J. Fluids Struct.* 26, 979–993.
- Amiralaei, M.R., Alighanbari, H., Hashemi, S.M., 2011. Flow field characteristics study of a flapping airfoil using computational fluid dynamics. *J. Fluids Struct.* 27, 1068–1085.
- Ashraf, M.A., Young, J., Lai, J.C.S., 2011. Reynolds number, thickness and camber effects on flapping airfoil propulsion. *J. Fluids Struct.* 27, 145–160.
- Broering, T.M., Lian, Y.S., 2015. Numerical study of tandem flapping wing aerodynamics in both two and three dimensions. *Comput. Fluids* 115, 124–139.
- Esfahani, J.A., Barati, E., Karbasian, H.R., 2015. Fluid structures of flapping airfoil with elliptical motion trajectory. *Comput. Fluids* 108, 142–155.
- Garrick, I.E., 1936. Propulsion of a flapping and oscillating airfoil. *NACA Rep.* 567, 1–9.
- Hoke, C.M., Young, J., Lai, J.C.S., 2015. Effects of time-varying camber deformation on flapping foil propulsion and power extraction. *J. Fluids Struct.* 56, 152–176.
- Jaworski, J.W., Gordnier, R.E., 2015. Thrust augmentation of flapping airfoils in low Reynolds number flow using a flexible membrane. *J. Fluids Struct.* 52, 199–209.
- Jones, K.D., Dohring, C.M., Platzer, M.F., 1998. Experimental and computational investigation of the Knoller-Betz effect. *AIAA J.* 36, 1240–1246.
- Lee, J.S., Lee, S.H., 2013. Fluid–structure interaction for the propulsive velocity of a flapping flexible plate at low Reynolds number. *Comput. Fluids* 71, 348–374.
- Lee, J.S., Seo, I.S., Lee, S.H., 2015. Propulsion velocity of a flapping wing at low Reynolds number. *J. Fluids Struct.* 54, 422–439.
- Lehmann, F.O., Pick, S., 2007. The aerodynamic benefit of wing–wing interaction depends on stroke trajectory in flapping insect wings. *J. Exp. Biol.* 210, 1362–1377.
- Licht, S.C., Wibawa, M.S., Hover, F.S., Triantafyllou, M.S., 2010. In-line motion causes high thrust and efficiency in flapping foils that use power downstroke. *J. Exp. Biol.* 213, 63–71.
- Lu, K., Xie, Y.H., Zhang, D., 2013. Numerical study of large amplitude, nonsinusoidal motion and camber effects on pitching airfoil propulsion. *J. Fluids Struct.* 36, 184–194.
- Mantia, M.L., Dabnichki, P., 2012. Added mass effect on flapping foil. *Eng. Anal. Bound. Elem.* 36, 579–590.
- Mivehchi, A., Dahl, J., Licht, S., 2016. Heaving and pitching oscillating foil propulsion in ground effect. *J. Fluids Struct.* 63, 174–187.
- Rozhdestvensky, K.V., Ryzhov, V.A., 2003. Aerohydrodynamics of flapping-wing propulsors. *Prog. Aerosp. Sci.* 39, 585–633.
- Sarkar, S., Chajjed, S., Krishnan, A., 2013. Study of asymmetric hovering in flapping flight. *Eur. J. Mech. B-Fluids* 37, 72–89.
- Sarkar, S., Venkatraman, K., 2005. Numerical simulation of incompressible viscous flow past a heaving airfoil. *Internat. J. Numer. Methods Fluids* 51, 1–29.
- Tian, F.B., Lu, X.Y., Luo, H.X., 2012. Propulsive performance of a body with a traveling-wave surface. *Phys. Rev. E* 86, 1–5.
- Tian, F.B., Luo, H.X., Song, J.L., Lu, X.Y., 2013. Force production and asymmetric deformation of a flexible flapping wing in forward flight. *J. Fluids Struct.* 36, 149–161.
- Tian, W., Bodling, A., Liu, H., Wu, J.C., He, G.W., Hu, H., 2016. An experimental study of the effects of pitch-pivot-point location on the propulsion performance of a pitching airfoil. *J. Fluids Struct.* 60, 130–142.
- Triantafyllou, M.S., Techet, A.H., Hover, F.S., 2004. Review of experimental work in biomimetic foils. *IEEE J. Oceanic Eng.* 29, 585–594.
- Wu, J., Chen, Y.L., Zhao, N., 2015a. Role of induced vortex interaction in a semi-active flapping foil based energy harvester. *Phys. Fluids* 27, 093601.
- Wu, J., Shu, C., 2009. Implicit velocity correction-based immersed boundary-lattice Boltzmann method and its applications. *J. Comput. Phys.* 228, 1963–1979.
- Wu, J., Yang, S.C., Shu, C., Zhao, N., Yan, W.W., 2015b. Ground effect on the power extraction performance of a flapping wing biomimetic energy generator. *J. Fluids Struct.* 54, 247–262.
- Young, J., Lai, J.C.S., 2004. Oscillation frequency and amplitude effects on the wake of a plunging airfoil. *AIAA J.* 42, 2042–2052.